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Semiconductor structure

Abstract

A semiconductor structure includes a first fin structure and a second fin structure, a first dielectric layer disposed over the first fin structure, a second dielectric layer disposed over the second fin structure, a first gate electrode disposed over the first dielectric layer, and a second gate electrode disposed over the second dielectric layer. A thickness of the first dielectric layer and a thickness of the second dielectric layer are equal. The second fin structure includes an outer region and an inner region, and a Ge concentration in the outer portion is less than Ge concentration in the inner portion.

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Background/Summary

PRIORITY DATA (1) This patent is a divisional application of U.S. patent application Ser. No. 17/330,272 filed May 25, 2021, which is a divisional application of U.S. patent application Ser. No. 16/373,235 filed Apr. 2, 2019, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/772,338 filed Nov. 28, 2018; each of these applications are incorporated herein by reference in their entirety.

BACKGROUND

(1) In the semiconductor art, it is desirable to improve transistor performance even as devices become smaller with ongoing reductions in scale. Strain-induced band structure modification and

mobility enhancement, which are used to increase drive current, are an attractive approach to improving transistor performance. For example, enhanced electron mobility in silicon would improve performance of an n-type metal-oxide-semiconductor (nMOS) device while enhanced hole mobility in silicon germanium (SiGe) would improve performance of a p-type MOS (pMOS) device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a flow diagram of some embodiments of a method of forming a dielectric layer.
- (3) FIGS. 2A to 2D are schematic drawings illustrating the method of forming the dielectric layer at various fabrication stages.
- (4) FIG. 3 is a flow diagram of some embodiments of a method of forming a semiconductor structure.
- (5) FIGS. 4A to 4F are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (6) FIG. 5 is a schematic drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.
- (7) FIGS. 6A to 6D are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (8) FIGS. 7A to 7H are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (9) FIG. 8 is a schematic drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.
- (10) FIG. 9 is a flow diagram of some embodiments of a method of forming a semiconductor structure.
- (11) FIGS. 10A to 10F are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (12) FIG. 11 is a schematic drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.
- (13) FIG. 12 is a drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.
- (14) FIGS. 13A to 13C are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (15) FIGS. 14A to 14H are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.
- (16) FIG. 15 is a schematic drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.
- (17) FIG. 16 is a schematic drawing illustrating a semiconductor structure at a fabrication stage

constructed according to aspects of the present disclosure in one or more embodiments.

(18) FIGS. 17A to 17F are cross-sectional views taken along a line I-I' of FIG. 16 illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments.

(19) FIG. 18 is a schematic drawing illustrating a semiconductor structure according to aspects of the present disclosure in one or more embodiments.

DETAILED DESCRIPTION

(20) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature on or over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of brevity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(21) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(22) As used herein, the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

(23) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins.

(24) SiGe is a semiconductor material that has a band gap smaller than that of silicon and can be controlled by varying Ge content. SiGe used in combination with silicon produces a heterojunction that provides low junction leakage and high mobility. In some embodiments, metal oxide semiconductor field effect transistor (MOSFET) devices have a SiGe channel that extends between a source region and a drain region. A gate electrode, configured to control the flow of charge carriers from the source region to the drain region, is separated from the SiGe channel by a gate dielectric layer. It is found that when the SiGe channel and the gate dielectric layer abut one another, Ge atoms may diffuse into the gate dielectric layer from the SiGe channel layer. Consequently, gate leakage current (J_g) is increased and reliability is reduced.

(25) To mitigate this problem different approaches are developed. For example, a thicker gate dielectric layer is formed on the SiGe channel layer in some comparative embodiments, but a SiGe

loss issue is raised. Further, the oxidation of the SiGe layer will form a silicon germanium oxide layer having a high interface trap density (Dit) that captures mobile charge carriers and results in low mobility. In other comparative embodiments, a thick gate dielectric layer is deposited on the SiGe channel layer, but such approach suffers from high interface state density. It is found that the interface between the deposited gate dielectric layer and the SiGe layer generally has unsatisfied bonds that act as interface charging centers, causing the “interface states”. The high density of such interface states indicates low quality of the deposited gate dielectric material and results in carrier mobility degradation.

(26) In still other comparative embodiments, a thin silicon cap layer may be formed between the SiGe channel layer and the gate dielectric layer. The silicon cap layer prevents diffusion of Ge atoms from the SiGe channel layer to the gate dielectric layer. However, it is found that if the silicon cap layer is not formed to an optimum thickness, the benefits of the SiGe channel layer are reduced. For example, when the silicon cap layer is too thin, Ge atoms may be able to diffuse into the silicon cap layer, and thus the silicon cap layer is transformed into a SiGe layer. The silicon cap layer therefore provides no function. Alternatively, if the silicon cap layer is too thick, the silicon cap layer becomes a part of the channel resulting in a high effective oxide thickness (EOT) and partial or full carrier spill over the silicon cap layer that reduces mobility. In advanced technology nodes, even a silicon cap layer formed to meet the optimal thickness is not able to meet the balance between an EOT scaling (e.g., below 1 nm) and a high mobility. Additionally, the use of the silicon cap layer increases process cost.

(27) Accordingly, the present disclosure provides a cap-free design for a gate dielectric layer on a SiGe channel layer. According to the cap-free design of the present disclosure, a dielectric layer is formed over a SiGe layer where a channel is to be formed, and a sacrificial semiconductor layer is formed on the dielectric layer. An anneal is subsequently performed. During the anneal, Ge atoms may diffuse from the SiGe layer, pass the first dielectric layer, and stay in the sacrificial semiconductor layer. The sacrificial semiconductor layer including the Ge atoms is then removed. Further, an interface between the SiGe layer and the dielectric layer can be improved during the anneal. Consequently, a dielectric layer having low interface trap density is obtained without SiGe loss. The dielectric layer used in a pMOS device having the SiGe layer where the channel is to be formed reduces gate leakage current and thus improves reliability. Accordingly, the cap-free design for the gate dielectric layer on a SiGe channel layer offers an opportunity to achieve high mobility and improve transistor performance.

(28) It should be noted that the cap-free dielectric design can be integrated in planar transistor devices and non-planar transistor devices, such as tri-gate, FinFET and gall-all-around (GAA) architectures. It should also be noted that the present disclosure presents embodiments in the form of multi-gate transistors or fin-type multi-gate transistors referred to herein as FinFET devices. The FinFET devices may be GAA devices, Omega-gate (a-gate) devices, Pi-gate (H-gate) devices, dual-gate devices, tri-gate devices, bulk devices, silicon-on-insulator (SOI) devices, and/or other configurations. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure.

(29) Further, the cap-free dielectric design of the present disclosure can also be integrated in gate-last approach or a replacement-gate (RPG) approach.

(30) FIG. 1 is a flow diagram of some embodiments of a method of forming a dielectric layer, and FIGS. 2A to 2D are schematic drawings illustrating the method of forming the dielectric layer at various fabrication stages. In some embodiments, a method of forming a dielectric layer **10** is provided. The method **10** includes a number of operations (**11**, **12**, **13**, **14**, **15** and **16**).

(31) Referring to FIGS. 1 and 2A, a substrate including a semiconductor layer **100** including a germanium (Ge) compound is received in operation **11**. In some embodiments, the semiconductor layer **100** includes at least two semiconductor materials having different lattice constants. For example but not limited thereto, the semiconductor layer **100** may include silicon germanium

(Si.sub.1-xGe.sub.x), wherein the germanium content, x, ranges from 0 to 1. In some embodiments, the Ge content may be greater than 0.3, but the disclosure is not limited thereto. The Ge content in the semiconductor layer **100** will be discussed in more detail in the following description. In some embodiments, the semiconductor layer **100** may include gallium arsenide (GaAs), indium phosphide (InP), aluminum gallium arsenide (AlGaAs), indium arsenide (InAs), or any other similar III-V material. In some embodiments, the semiconductor layer **100** is formed within the substrate. In some embodiments, at least a fin structure is disposed over the substrate and protrudes from the substrate. Further, the fin structure includes the semiconductor layer **100**. In some embodiments, a plurality of nanowires is disposed over the substrate. Further, each nanowire includes the semiconductor layer **100**.

(32) In operation **12**, a dielectric layer **110** is formed on the semiconductor layer **100**. In some embodiments, the dielectric layer **110** can include a semiconductor oxide. For example, the dielectric layer **110** may include a silicon oxide (SiO.sub.x) layer, such as a silicon dioxide (SiO.sub.2) layer, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **110** may include a first dielectric layer **112a** and a second dielectric layer **112b**. As shown in FIG. 2A, the first dielectric layer **112a** is sandwiched between the second dielectric layer **112b** and the semiconductor layer **100**. In some embodiments, the first dielectric layer **112a** may be a III-V compound semiconductor oxide layer. For example but not limited thereto, the first dielectric layer **112a** may be a silicon germanium oxide (Si.sub.xGe.sub.yO) layer, wherein x is between approximately 0.6 and approximately 1, and y is between approximately 0.4 and approximately 0. In some embodiments, the III-V compound semiconductor oxide layer may be a native oxide layer that is spontaneously grown on a surface of the semiconductor layer **100**. For example, the first dielectric layer **112a** may be a native silicon germanium oxide layer spontaneously grown on a surface of the semiconductor layer **100**. The second dielectric layer **112b** may be a silicon oxide layer. In some embodiments, the second dielectric layer **112b** (i.e., the silicon oxide layer) may be a chemical oxide layer, and the second dielectric layer **112b** can be formed by suitable thermal oxidation or deposition. In some embodiments, the second dielectric layer **112b** can be formed by ozone (O.sub.3) and H.sub.2O. In some embodiments, the second dielectric layer **112b** can be formed by an ozone assisted sub-atmospheric pressure thermal chemical vapor deposition (SACVD), but the disclosure is not limited thereto. In some embodiments, the second dielectric layer **112b** can be formed by H.sub.2SO.sub.4 and H.sub.2O. In some embodiments, both of the first and second dielectric layers **112a** and **112b** can be formed by O.sub.3 and H.sub.2O or by H.sub.2SO.sub.4 and H.sub.2O, but the disclosure is not limited thereto. In some embodiments, the second dielectric layer **112b** can be formed by thermal oxidation and deposition. For example, the second dielectric layer **112b** can be formed by a thermal oxidation and a plasma-enhanced atomic layer deposition (PEALD). In some embodiments, both of the first and second dielectric layers **112a** and **112b** can be formed by thermal oxidation and deposition. In some embodiments, bis(diethylamino)silane (SiH.sub.2(NEt.sub.2).sub.2, SAM 24), N,N-diisopropylaminosilane (DIPAS, LTO520), tetrakis-dimethylaminosilane (SiH(NMe.sub.2).sub.3, TDMAS), Si.sub.2C.sub.16, silane (SiH.sub.4), disilane (Si.sub.2H.sub.6), nitrogen (N.sub.2), oxygen (O.sub.2), nitrous oxide (N.sub.2O) and ozone can be used in the PEALD. In some embodiments, a thickness of the dielectric layer **110** is between approximately 10 angstroms (Å) and approximately 50 Å, but the disclosure is not limited thereto. In other embodiments, a thickness of the dielectric layer **110** is less than 20 Å, but the disclosure is not limited thereto.

(33) Referring to FIGS. 1 and 2B, a first sacrificial cap **120** including silicon is formed on the dielectric layer **110** in operation **13**. In some embodiments, the first sacrificial cap **120** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the first sacrificial cap **120** is greater than 20 Å, but the disclosure is not limited thereto. In some embodiments, the first sacrificial cap **120** can be formed by silane, disilane, trisilane (Si.sub.3H.sub.8), LTO520, tetrasilane (Si.sub.4H.sub.10) and N.sub.2, but the disclosure is not

limited thereto.

(34) Referring to FIGS. 1 and 2C, the substrate is annealed to transform the first sacrificial cap **120** into a second sacrificial cap **120'** including SiGe in operation **14**. In some embodiments, an anneal **130** is performed with N.sub.2, hydrogen (H.sub.2), argon (Ar) and oxygen (O.sub.2). In some embodiments, the anneal **130** is performed at a temperature between approximately 300° C. and approximately 1100° C., but the disclosure not limited thereto. During the anneal **130**, Ge atoms diffuse from the semiconductor layer **100** and from the first dielectric layer **112a**. Further, Ge atoms may diffuse upwardly and pass the second dielectric layer **112b**, and stay with Si atoms in the first sacrificial cap **120**. Accordingly, the first sacrificial cap **120** including silicon is transformed into the second sacrificial cap **120'** including silicon germanium. The second sacrificial cap **120'** may include Ge atoms from the semiconductor layer **100** and from the first dielectric layer **112a**. It should be noted that the thickness of the first sacrificial cap **120** (now the second sacrificial cap **120'**) is greater than 20 Å in order to provide sufficient accommodations for the Ge atoms.

(35) As described above, Ge atoms may diffuse from the semiconductor layer **100**. Accordingly, a portion of the semiconductor layer **100** may lose Ge atoms. In some embodiments, it is observed that a portion, usually an upper portion, of the semiconductor layer **100** is likely to exhibit Ge diffusion. Accordingly, the portion having a lower Ge concentration is recognized and defined as a first portion **100U** while the other portion, usually a portion lower than the first portion **100U**, having the Ge concentration greater than that of the first portion **100U**, is recognized and defined as a second portion **100L**. In some embodiments, the Ge concentration in the second portion **100L** may be substantially equal to an original Ge concentration in the semiconductor layer **100**. The first portion **100U** of the semiconductor layer **100** has a first Ge concentration prior to annealing of the substrate and a second Ge concentration after the annealing of the substrate. In some embodiments, the first Ge concentration in the first portion **100U** is substantially equal to the Ge concentration in the second portion **100L**, which is the original Ge concentration. The second Ge concentration in the first portion **100U** is lower than the first Ge concentration in the first portion **100U** and the Ge concentration in the second portion **100L**. For example but not limited thereto, the first Ge concentration in the first portion **100U** and the Ge concentration in the second portion **100L** may be greater than approximately 30%, while the second Ge concentration in the first portion **100U** may be less than approximately 25%. Accordingly, a Ge concentration in an interface **140** between the dielectric layer **110** and the semiconductor layer **100** is reduced from greater than approximately 30% to less than approximately 25%, but the disclosure is not limited thereto.

(36) Further, Ge atoms in the first dielectric layer **112a** may also diffuse into the first sacrificial cap **120**, and thus the first dielectric layer **112a**, which includes a silicon germanium oxide layer, may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **110'** including silicon oxide may be formed after operation **14**. In some embodiments, the dielectric layer **110'** includes the original second dielectric layer **112b** including silicon oxide and the first dielectric layer **112a** previously including silicon germanium oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **110'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **110'** is less than 1.5%.

(37) Referring to FIGS. 1 and 2D, the second sacrificial cap **120'** is removed to expose the dielectric layer **110'** in operation **15**. In some embodiments, a gate electrode is formed over the dielectric layer **110'** in operation **16**. In some embodiments, the dielectric layer **110'** serves as a gate dielectric layer for a transistor device, and the thickness of the dielectric layer **110'** may be between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In such embodiments, the gate electrode can include a semiconductor material. In other embodiments, the dielectric layer **110'** serves as an interfacial layer (IL) in an RPG approach, and the thickness of the dielectric layer **110'** is less than 20 Å, but the disclosure is not limited thereto. In such embodiments, the gate electrode can include metal materials, and a high-k gate dielectric layer is sandwiched between the dielectric layer **110'** and the metal gate electrode. The operation **16** will be

discussed in more detail in the following description.

(38) According to the method of forming the dielectric layer **10**, the first sacrificial cap **120** is formed to provide accommodations for Ge atoms diffused from the dielectric layer (i.e., the first dielectric layer **112a**) including silicon germanium oxide and from the semiconductor layer **100** in the anneal **130**. Further, because the second sacrificial cap **120'** (transformed from the first sacrificial cap **120**) is removed, the method **10** provides a cap-free dielectric design. According to the method **10**, the Ge concentration in the dielectric layer **110'** is less than 3% or even less than 1.5%. Consequently, gate leakage current (J_g) is reduced by the dielectric layer **110'** and thus device reliability is improved.

(39) Further, there are two interface issues causing carrier mobility degradation as mentioned in reference to the comparative embodiments: interface trap density (D_{it}) caused by SiGe oxidation and high density of interface states caused by dielectric deposition. The SiGe oxidation will form a silicon germanium oxide layer having a high interface trap density (D_{it}) that captures mobile charge carriers and results in low mobility, while the high density of interface states indicates low quality of the deposited gate dielectric material and results in carrier mobility degradation. Both of the two interface issues are mitigated by the method **10**. It should be noted that because there are at least three layers (i.e., the first sacrificial cap **120**, the second dielectric layer **112b** and the first dielectric layer **110**) formed over the semiconductor layer **100** including a Ge compound, oxygen can be blocked by the three layers during the forming of the second dielectric layer **112b** and/or the annealing of the first sacrificial cap **120**. In other words, SiGe oxidation in the semiconductor layer **100** can be avoided by the three layers. Accordingly, interface trap density (D_{it}) caused by SiGe oxidation is reduced. Further, the interface **140** is formed between the dielectric layer **110'** and the semiconductor layer **100**. Specifically, the interface **140** is formed between the previously first dielectric layer **112a** (now a portion of the dielectric layer **110'**) and the semiconductor layer **100**. As described above, the first dielectric layer **112a** may be a native oxide layer, instead of a chemical oxide layer that is formed by deposition. Therefore, the density of interface states, which is created by deposition, is reduced. Accordingly, carrier mobility degradation is mitigated.

(40) In some embodiments, the cap-free dielectric design described above can be integrated in planar transistor devices. FIG. 3 is a flow diagram of some embodiments of a method of forming a semiconductor structure, and FIGS. 4A to 4F are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments. In some embodiments, a method of forming a semiconductor structure **30** is provided, and the method **30** includes a number of operations (**311**, **312**, **313**, **314**, **315**, **316** and **317**).

(41) Referring to FIGS. 3 and 4A, a substrate **400** including a channel layer **402** is received in operation **311**. The substrate **400** can include a bulk silicon substrate, a single crystalline silicon substrate (doped or undoped), or a semiconductor-on-insulator (SOI) substrate. In some embodiments, the substrate **400** may have a doping type (e.g., an n-type doping). In some embodiments, the substrate **400** may include a doped epitaxial layer disposed on a semiconductor body including a bulk silicon. In some embodiments, isolation structures (not shown) such as shallow trench isolation (STI) structures may be formed in the substrate **400** to define regions where devices to be formed and to electrically isolate the devices to be formed.

(42) Still referring to FIG. 4A, in some embodiments, a SiGe MOSFET device is to be formed over the substrate **400** to take advantage of the low junction leakage and high mobility provided by SiGe and Si hetero-junction. Accordingly, a semiconductor layer is formed to serve as the channel layer **402**. The channel layer **402** may include a Ge compound. For example, the channel layer **402** may include $\text{Si}_{1-x}\text{Ge}_x$ alloy, wherein the germanium content, x , ranges from 0 to 1. In some embodiments, the germanium content may be greater than 0.3, but the disclosure is not limited thereto. In some embodiments, the Ge content may be greater than 0.35, but the disclosure is not limited thereto. In other embodiments, the channel layer **402** can include other III-V semiconductor

materials having alloy including a combination of group III material (i.e., group 13 in the periodic table) and group V material (i.e., group 15 in the periodic table). For example, in some embodiments, the channel layer **402** can include GaAs, InP, AlGaAs, InAs, or any other similar material.

(43) Referring to FIGS. **3** and **4B**, a dielectric layer **410** is formed on the channel layer **402** in operation **312**. The dielectric layer **410** can include a semiconductor oxide. For example, the dielectric layer **410** may include a silicon oxide layer, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **410** may include a first dielectric layer **412a** and a second dielectric layer **412b**. As shown in FIG. **4B**, the first dielectric layer **412a** is sandwiched between the second dielectric layer **412b** and the channel layer **402**. In some embodiments, the first dielectric layer **412a** may be a III-V compound semiconductor oxide layer. For example but not limited thereto, the first dielectric layer **412a** may be a Si.sub.xGe.sub.yO layer, wherein x is between approximately 0.6 and approximately 1, and y is between approximately 0.4 and approximately 0. In some embodiments, the III-V compound semiconductor oxide layer may be a native oxide layer that is spontaneously grown on a surface of the channel layer **402**. For example, the first dielectric layer **412a** may be a native silicon germanium oxide layer spontaneously grown on a surface of the channel layer **402**. In some embodiments, the second dielectric layer **412b** may include a silicon oxide layer. In some embodiments, the second dielectric layer **412b** may be a chemical oxide layer, and the second dielectric layer **412b** can be formed by suitable thermal oxidation or deposition. The method of forming the second dielectric layer **412b** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, a thickness of the dielectric layer **410** is between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In other embodiments, the thickness of the dielectric layer **410** is less than 20 Å, but the disclosure is not limited thereto.

(44) Still referring to FIGS. **3** and **4B**, a first sacrificial cap **420** including Si is formed on the dielectric layer **110** in operation **313**. In some embodiments, the first sacrificial cap **420** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the first sacrificial cap **420** is greater than 20 Å, but the disclosure is not limited thereto. The method of forming the first sacrificial cap **420** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity.

(45) Referring to FIGS. **3** and **4C**, the substrate **400** is annealed to transform the first sacrificial cap **420** into a second sacrificial cap **420'** including SiGe in operation **314**. In some embodiments, an anneal **430** is performed, wherein the details of the anneal **430** can be similar to those of anneals described above; therefore, similar details are omitted in the interest of brevity. During the anneal **430**, Ge atoms diffuse from the channel layer **402** and from the first dielectric layer **412a**. Further, Ge atoms may diffuse upwardly and pass the second dielectric layer **412b**, and stay with Si atoms in the first sacrificial cap **420**. Accordingly, the first sacrificial cap **420** including silicon is transformed into the second sacrificial cap **420'** including silicon germanium. In other words, the second sacrificial cap **420'** may include Ge atoms from the channel layer **402** and from the first dielectric layer **412a**. It should be noted that the thickness of the first sacrificial cap **420** (now the second sacrificial cap **420'**) is greater than 20 Å in order to provide sufficient accommodations for Ge atoms.

(46) As described above, Ge atoms may diffuse from the channel layer **402**. Accordingly, a portion of the semiconductor layer **402** may lose Ge atoms. In some embodiments, it is observed that a portion of the channel layer **402**, usually an upper portion, is likely to exhibit Ge diffusion. Accordingly, the portion having lower Ge concentration is recognized and defined as a first portion **402U** while the other portion, usually a portion lower than the first portion **402U**, having the Ge concentration greater than that of the first portion **402U** is recognized and defined as a second portion **402L**. In some embodiments, the Ge concentration in the second portion **402L** may be substantially equal to an original Ge concentration in the channel layer **402**. The first portion **402U**

of the channel layer **402** has a first Ge concentration prior to the annealing of the substrate **400** and a second Ge concentration after the annealing of the substrate **400**. In some embodiments, the first Ge concentration in the first portion **402U** is substantially equal to the Ge concentration in the second portion **402L**, which is the original Ge concentration. The second Ge concentration in the first portion **402U** is lower than the first Ge concentration in the first portion **402U** and the Ge concentration in the second portion **402L**. For example but not limited thereto, the first Ge concentration in the first portion **402U** and the Ge concentration in the second portion **402L** may be greater than approximately 30%, while the second Ge concentration in the first portion **402U** may be less than approximately 25%. Accordingly, a Ge concentration in an interface **440** between the dielectric layer **410** and the channel layer **402** is reduced from greater than approximately 30% to less than approximately 25%, but the disclosure is not limited thereto.

(47) Further, Ge atoms in the first dielectric layer **412a** may also diffuse into the first sacrificial cap **420**, and thus the first dielectric layer **412a**, which includes a silicon germanium oxide layer, may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **410'** including silicon oxide may be formed after operation **314**. In some embodiments, the dielectric layer **410'** includes the original second dielectric layer **412b** including silicon oxide and the first dielectric layer **412a** previously including silicon germanium oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **410'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **410'** is less than 1.5%.

(48) Referring to FIGS. **3** and **4D**, the second sacrificial cap **420'** is removed to expose the dielectric layer **410'** in operation **315**. Referring to FIGS. **3** and **4E**, a semiconductor gate layer **450** is formed over the dielectric layer **410'** in operation **316**. In some embodiments, the dielectric layer **410'** serves as a gate dielectric layer for a transistor device, and a thickness of the dielectric layer **410'** may be between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In such embodiments, the gate layer can include a semiconductor material; for example, a doped polysilicon is formed to serve as the semiconductor gate layer **450**.

(49) Referring to FIG. **4F**, in some embodiments, a patterned hard mask **452** is formed on the semiconductor gate layer **450**, and the semiconductor gate layer **450** is patterned through the patterned hard mask **452**, such that a gate structure including the semiconductor gate layer **450** and the dielectric layer **410'** is obtained. In some embodiments, source/drain extension regions **460** can be formed in the substrate **400** at two sides of the gate structure, spacers **462** can be formed on sidewalls of the gate structure, and a source/drain **464** can be formed in operation **317**. The source/drain **464** is formed in the substrate **400** at two sides of the gate structure and the spacers **462**. In some embodiments, the source/drain **464** can be a strained source/drain (S/D) structure. The strained S/D structures **464** can be formed by growing a strained material in recesses (not shown) by an epitaxial (epi) operation. In some embodiments, a lattice constant of the strained material may be different from a lattice constant of the substrate **400**. In some embodiments, the strained S/D structures **464** may include Ge, SiGe, InAs, InGaAs, InSb, GaSb, InAlP, InP, or a combination thereof, but the disclosure is not limited thereto. Accordingly, a semiconductor structure such as a planar SiGe MOSFET device **40** is obtained, as shown in FIG. **4F**.

(50) The method of forming the semiconductor structure **30** uses a cap-free dielectric design. According to the method **30**, the Ge concentration in the dielectric layer **410'** is less than 3% or even less than 1.5%. Consequently, gate leakage current (J_g) is reduced by the dielectric layer **410'** and reliability of the SiGe MOSFET device **40** is improved. Further, interface trap density (D_{it}) and density of interface states are also reduced by the method **30**. Therefore, the carrier mobility is improved by the SiGe channel layer **402** while the carrier mobility degradation issue caused by high D_{it} and density of interface states is mitigated.

(51) In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally

provides benefits by increasing production efficiency and lowering associated costs. MOSFET devices have typically been formed with a gate dielectric layer including SiO and a gate electrode including polysilicon, as described above. There has been a desire to replace the SiO gate dielectric layer and polysilicon gate electrode with a high-k gate dielectric and a metal gate electrode to improve device performance as feature sizes continue to decrease. In some embodiments, the method **30** can further include a number of operations (**321**, **322**, **323**, **324**, **325**, **326**, **327** and **328**). In some embodiments, operation **321** can be performed after operation **317**, but the disclosure is not limited thereto. In some embodiments, operations **327** and **328** can be performed after operation **321**, but the disclosure is not limited thereto. In other embodiments, operation **322**, **323**, **324**, **325**, **326**, **327** and **328** can be performed after the operation **321**, but the disclosure is not limited thereto. (52) Further, as the integrated circuit size continues to shrink, a core operation voltage is reduced. It is expected that the core operation voltage will continue to be reduced as the integrated circuit size continues to shrink. As the core operation voltage is reduced, an I/O operation voltage stays at a higher value. As a result, the MOS devices in an I/O region and a core region are expected to work under different operation voltages. Thus different operations can be performed to form the devices in the I/O region and the core region. For example, in some embodiments, the above-mentioned operations **311**, **312**, **313**, **314**, **315** and **316** are performed to simultaneously form a device **40a** having a polysilicon gate electrode **450** in an I/O region **404a** and a device **40b** having polysilicon gate electrode **450** in a core region **404b**, as shown in FIG. 5.

(53) In some embodiments, a dielectric structure **470** is formed over the substrate **400**. In some embodiments, the dielectric structure **470** can include an etch-stop layer (e.g., a contact etch stop layer (CESL)) **472** and various dielectric layers (e.g., an inter-layer dielectric (ILD) layer) **474** formed over the substrate **400** after the forming of the strained S/D structures **464**. In some embodiments, the CESL **472** includes a SiN layer, a SiCN layer, a SiON layer, and/or other materials known in the art. In some embodiments, the ILD layer **474** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. Accordingly, the SiGe MOSFET device **40a** and the SiGe MOSFET device **40b** are embedded in the dielectric structure **470**.

(54) In some embodiments, the operations **321**, **327** and **328** can be performed to form a MOSFET device **40a** having a metal gate electrode in the I/O region **404a**, but the disclosure is not limited thereto. Accordingly, only the device **40a** in the I/O region **404a** is shown in FIGS. 6A to 6D. In some embodiments, the semiconductor gate layer **450** (i.e., the polysilicon gate electrode) serves as a sacrificial gate structure, also known as a dummy gate structure, in a replacement gate (RPG) approach. Referring to FIGS. 3 and 6A, in some embodiments, after the CESL **472** and the ILD layer **474** are deposited, a planarization process, such as a chemical mechanical planarization (CMP) operation, may be performed to remove a portion of the dielectric structure **470**, a portion of the spacer **462** and the patterned hard mask **452**. Consequently, a top surface of the semiconductor gate structure **450** is exposed, as shown in FIG. 6A.

(55) Referring to FIGS. 3 and 6B, the semiconductor gate structure **450** is removed in operation **321**. Consequently, a gate trench **454** is formed within the dielectric structure **470** and the spacers **462**. Further, the dielectric layer **410'** is exposed through a bottom of the gate trench **454**.

(56) Referring to FIGS. 3 and 6C, a high-k dielectric layer **480** is formed on the dielectric layer **410'** in operation **327**. In some embodiments, the high-k dielectric layer **480** may include hafnium oxide (HfO₂), zirconium oxide (ZrO₂), lanthanum oxide (La₂O₃), aluminum oxide (Al₂O₃), titanium oxide (TiO₂), yttrium oxide (Y₂O₃), strontium titanate (SrTiO₃), hafnium oxynitride (HfO_xNy), other suitable metal-oxides, or combinations thereof. Additionally, the dielectric layer **410'** serves as an interfacial layer (IL) between the high-k dielectric layer **480** and the channel layer **402**.

(57) Referring to FIGS. 3 and 6D, a metal gate structure **482** is formed over the high-k dielectric layer **480** in operation **328**. In some embodiments, the metal gate structure **482** can include at least a barrier metal layer (not shown), a work functional metal layer **484** and a gap-filling metal layer **486**. The barrier metal layer can include, for example but not limited to, TiN. The work function metal layer **484** can include a single layer of TiN, TaN, TaAlC, TiC, TaC, Co, Al, TiAl, HfTi, TiSi, TaSi or TiAlC, or a multilayer of two or more of these materials, but is not limited thereto. For an n-channel FET, one or more of TaN, TaAlC, TiN, TiC, Co, TiAl, HfTi, TiSi and TaSi are used as the work function metal layer **484**, and for a p-channel FET, one or more of TiAlC, Al, TiAl, TaN, TaAlC, TiN, TiC and Co are used as the work function metal layer **484**. In some embodiments, the gap-filling metal layer **486** can include conductive material such as Al, Cu, AlCu, or W, but the material is not limited thereto.

(58) Accordingly, a planar SiGe MOSFET **40a** having a metal gate electrode is obtained in the I/O region **404a**, according to the method **30**.

(59) In contrast to the device in the I/O region **404a**, the device **40b** having a metal gate electrode in the core region **404b** can be formed by the operations **321**, **322**, **323**, **324**, **325**, **326**, **327** and **328**, but the disclosure is not limited thereto. Accordingly, only the device **40b** in the core region **404b** is shown in FIGS. 7A to 7H. In some embodiments, the semiconductor gate layer **450** (i.e., the polysilicon gate electrode) serves as a sacrificial gate, also known as a dummy gate, in a replacement gate (RPG) approach. Referring to FIGS. 3 and 7A, as described above, after the CESL **472** and the ILD layer **474** are deposited, a planarization process, such as a CMP operation, may be performed to remove a portion of the dielectric structure **470**, a portion of the spacer **462** and the patterned hard mask **452**. Consequently, a top surface of the semiconductor gate structure **450** is exposed. The semiconductor gate structure **450** is removed in operation **321**. Consequently, a gate trench **454** is formed within the dielectric layer **470** and the spacers **462**. Further, the dielectric layer **410'** is exposed through a bottom of the gate trench **454**.

(60) Referring to FIGS. 3 and 7B, the dielectric layer **410'** is removed in operation **322**. Consequently, the channel layer **402**, such as the first portion **402U**, is exposed through the bottom of the gate trench **454**.

(61) Referring to FIGS. 3 and 7C, another dielectric layer **414** is formed on the channel layer **402** in operation **323**. The dielectric layer **414** can include semiconductor oxide. For example, the dielectric layer **414** may include a silicon oxide layer, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **414** may include a first dielectric layer **416a** and a second dielectric layer **416b**. As shown in FIG. 7C, the first dielectric layer **416a** is sandwiched between the second dielectric layer **416b** and the channel layer **402**. In some embodiments, the first dielectric layer **416a** may be a III-V compound semiconductor oxide layer and the second dielectric layer **416b** may include a silicon oxide layer. For example but not limited thereto, the first dielectric layer **416a** may be a silicon germanium oxide ($\text{Si}_{1-x}\text{Ge}_x\text{O}$) layer wherein x is between approximately 0.6 and approximately 1, and y is between approximately 0.4 and approximately 0. In some embodiments, the III-V compound semiconductor oxide layer may be a native oxide layer that is spontaneously grown on a surface of the channel layer **402**. For example, the first dielectric layer **416a** may be a native silicon germanium oxide layer spontaneously grown on a surface of the channel layer **402**. In some embodiments, the second dielectric layer **416b** may be a silicon oxide layer, and may be a chemical oxide layer. The second dielectric layer **416b** can be formed by suitable thermal oxidation or deposition. The method of forming the second dielectric layer **416b** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, a thickness of the dielectric layer **414** is less than approximately 20 Å, but the disclosure is not limited thereto. In some embodiments, when the second dielectric layer **416b** of the dielectric layer **414** is formed by deposition, the second dielectric layer **416b** may cover sidewalls of the gate trench **454** and top surfaces of the spacers **462** and the dielectric structure **470**, as shown in FIG. 7C, but the disclosure is not limited thereto.

(62) Referring to FIGS. 3 and 7D, a third sacrificial cap **422** including Si is formed on the dielectric layer **414** in operation **324**. In some embodiments, the third sacrificial cap **422** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the third sacrificial cap **422** is greater than 10 Å, but the disclosure is not limited thereto. The method of forming the third sacrificial cap **422** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity.

(63) Referring to FIGS. 3 and 7E, the substrate **400** is annealed to transform the third sacrificial cap **422** into a fourth sacrificial cap **422'** in operation **325**. In some embodiments, an anneal **432** is performed, wherein the details of the anneal **432** can be similar to those of anneals described above; therefore, similar details are omitted in the interest of brevity. During the anneal **432**, Ge atoms may diffuse from the channel layer **402** and from the first dielectric layer **416a**. Further, the Ge atoms may diffuse upwardly and pass the second dielectric layer **416b**, and stay with Si atoms in the third sacrificial cap **422**. Accordingly, the third sacrificial cap **422** including Si is transformed into the fourth sacrificial cap **422'** including silicon germanium. The fourth sacrificial cap **422'** may include Ge atoms from the channel layer **402** and from the first dielectric layer **416a**. It should be noted that the thickness of the third sacrificial cap **422** (now the fourth sacrificial cap **422'**) is greater than 10 Å in order to provide sufficient accommodations for Ge atoms.

(64) It should be noted that the first portion **402U'** of the channel layer **402** in the core region **404b** may undergo the anneal twice (i.e., the anneal **430** and the anneal **432**), therefore more Ge atoms may diffuse from the first portion **402U'** of the channel layer **402** in the core region **404b**. Accordingly, the Ge concentration in the first portion **402U'** of the channel layer **402** in the core region **404b** may be further reduced.

(65) Further, Ge atoms in the first dielectric layer **416a** may also diffuse into the third sacrificial cap **422**, and thus the first dielectric layer **416a**, which includes a silicon germanium oxide layer, may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **414'** including silicon oxide may be formed after operation **325**. In some embodiments, the dielectric layer **414'** includes the original second dielectric layer **416b** including silicon oxide and the first dielectric layer **416a** previously including silicon germanium oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **414'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **414'** is less than 1.5%.

(66) Referring to FIGS. 3 and 7F, the fourth sacrificial cap **422'** is removed to expose the dielectric layer **414'** in operation **326**. In some embodiments, the dielectric layer **414'** serves as an IL for a transistor device, and a thickness of the dielectric layer **414'** may be less than approximately 20 Å, but the disclosure is not limited thereto.

(67) Referring to FIGS. 3 and 7G, a high-k dielectric layer **480** is formed on the dielectric layer **414'** in operation **327**. In some embodiments, the high-k dielectric layer **480** may include HfO₂, ZrO₂, La₂O₃, Al₂O₃, TiO₂, Y₂O₃, SrTiO₃, HfO_xNy, other suitable metal-oxides, or combinations thereof.

(68) Referring to FIGS. 3 and 7H, a metal gate structure **482** is formed over the high-k dielectric layer **480** in operation **328**. In some embodiments, the metal gate structure **482** can include at least a barrier metal layer (not shown), a work function metal layer **484** and a gap-filling metal layer **486**. The barrier metal layer can include, for example but not limited to, TiN. The work function metal layer **484** can include a single layer of TiN, TaN, TaAlC, TiC, TaC, Co, Al, TiAl, HfTi, TiSi, TaSi or TiAlC, or a multilayer of two or more of these materials, but is not limited thereto. For an n-channel FET, one or more of TaN, TaAlC, TiN, TiC, Co, TiAl, HfTi, TiSi and TaSi are used as the work function metal layer **484**, and for a p-channel FET, one or more of TiAlC, Al, TiAl, TaN, TaAlC, TiN, TiC and Co are used as the work function metal layer **484**. In some embodiments, the gap-filling metal layer **486** can include conductive material such as Al, Cu, AlCu, or W, but is not limited to such materials.

(69) Accordingly, a planar SiGe MOSFET **40b** having a metal gate is obtained in the core region

404b, according to the method **30**.

(70) According to the method **30**, when the planar SiGe MOSFET device having the metal gate **40a** is required in the I/O region **404a**, operations **327** and **328** can be performed after operation **321**, and when the planar SiGe MOSFET device having the metal gate **40b** is required in the core region **404b**, operations **322** to operation **328** can be performed after operation **321**. In some embodiments, the semiconductor gate layer **450** in both of the device **40a** and **40b** can be removed simultaneously, in operation **321**, and a protecting layer (not shown) can be formed in the I/O region **404a** after the removal of the semiconductor gate layer in order to protect the dielectric layer **410'** in the device **40a** in the I/O region **404a**. The operations **322** to **326** can be performed on the device **40b** in the core region **404b** without impacting the device **40a** in the I/O region **404a**. In such embodiments, the protecting layer can be removed after the removal of the fourth sacrificial cap **422'** (i.e., operation **326**), and the high-k gate dielectric layer **480** and the metal gate layer **482** can be formed in the gate trenches **454** in the SiGe MOSFET device **40a** in the I/O region **404a** and in the MOSFET device **40b** in the core region **404b**.

(71) Referring to FIG. **8**, accordingly, a semiconductor structure is provided. The semiconductor structure includes a substrate **400** including a first region **404a** (i.e., an I/O region) and a second region **404b** (i.e., a core region). The semiconductor structure includes a channel layer **402** disposed in the first region **404a** and a channel layer **402** disposed in the second region **404b**. As described above, the channel layer **402** in the first region **404a** includes a Ge compound such as silicon germanium and the channel layer **402** in the second region **404b** includes a Ge compound such as silicon germanium. In some embodiments, the channel layer **402** in the first region **404a** includes a first portion **402U** and a second portion **402L**, wherein the Ge concentration in the first portion **402U** is less than that in the second portion **402L**. In some embodiments, the channel layer **402** in the second region **404b** includes a first portion **402U'** and the second portion **402L**, wherein the Ge concentration in the first portion **402U'** is less than that of the second portion **402L**. In some embodiments, the second portions **402L** of the channel layers **402** in both of the first and second regions **404a** and **404b** may include the same Ge concentration. However, because the first portion **402U'** of the channel layer **402** in the core region **404b** may undergo the anneal twice (i.e., the anneal **430** and the anneal **432**), more Ge atoms may diffuse from the first portion **402U'** of the channel layer **402** in the core region **404b**. Accordingly, the Ge concentration in the first portion **402U'** of the channel layer **402** in the second region **404b** may be less than the Ge concentration in the first portion **402U** of the channel layer **402** in the first region **404a**.

(72) Still referring to FIG. **8**, the semiconductor structure further includes a dielectric layer **410'** disposed on the channel layer **402** in the first region **404a** and a dielectric layer **414'** disposed on the channel layer **402** in the second region **404b**. In some embodiments, the dielectric layer **410'** and the dielectric layer **414'** may serve as an interfacial layer (IL), wherein a high-k dielectric layer **480** is disposed on the dielectric layer **410'** and the dielectric layer **414'**, respectively, as shown in FIG. **8**. A metal gate electrode including a work function metal layer **484** and a gap-filling metal layer **486** can be formed on the high-k dielectric layer **480**, respectively, as shown in FIG. **8**.

(73) The dielectric layer **410'** includes a thickness **T1**, and the dielectric layer **414'** includes a thickness **T2**. In some embodiments, the thickness **T1** of the dielectric layer **410'** in the first region **404a** is greater than the thickness **T2** of the dielectric layer **414'** in the second region **404b**. In some embodiments, the thickness **T1** of the dielectric layer **410'** in the first region **404a** is between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In some embodiments, the thickness **T2** of the dielectric layer **414'** in the second region **404b** is less than 20 Å, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **410'** in the first region **404a** may be disposed between the high-k dielectric layer **480** and the channel layer **402**, and may have a flat shape, as shown in FIG. **8**. In some embodiments, the dielectric layer **414'** in the second region **404b** may be disposed not only between the high-k dielectric layer **480** and the channel layer **402** but also between the high-k dielectric layer **480** and the spacer **462**. Further, the

dielectric layer **414'** may have a U shape, as shown in FIG. **8**. Additionally, portions of the dielectric layer **414'** are in contact with the spacer **462** and a portion of the dielectric layer **414'** is in contact with the channel layer **402**. In some embodiments, a thickness of the portions of the dielectric layer **414'** that are in contact with the spacer **462** is less than a thickness of the portion of the dielectric layer **414'** that is in contact with the channel layer **402**, but the disclosure is not limited thereto. However, in some embodiments, when the dielectric layer **414'** is formed by oxidation or thermal oxidation, the dielectric layer **414'** can have a flat shape, though not shown. (74) According to the method **30**, the Ge concentrations in the dielectric layers **410'** and **414'** are less than 3% or even less than 1.5%, while the Ge concentration in the dielectric layer **414'** may be even less than that in the dielectric layer **410'**. Consequently, gate leakage current (J_g) is reduced by the dielectric layers **410'** and **414'** and reliability of the SiGe MOSFET devices **40a** and **40b** is improved. Further, interface trap density (D_{it}) and density of interface states are also reduced by the method **30**. Therefore the carrier mobility of the SiGe MOSFET devices **40a** and **40b** is improved by the SiGe channel layer **402** while the carrier mobility degradation issue caused by high D_{it} and density of interface states is mitigated.

(75) As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance and lower costs, challenges from both fabrication and design issues have led to the development of three-dimensional designs, such as the FinFET device. FinFET devices are fabricated with a thin “fin” or “fin structure” vertically extending from a substrate, wherein a gate electrode is formed over the fin. Thus, the channel of the FinFET device is formed. In some embodiments, the cap-free dielectric design described above can be integrated in non-planar transistor devices, such as FinFET devices. The FinFET device, for example, may be a complementary metal-oxide-semiconductor (CMOS) device including a p-type MOS FinFET device and an n-type MOS FinFET device. The following disclosure will continue with description of a CMOS FinFET device to illustrate various embodiments of the present disclosure. It is understood, however, that the application should be not limited to a particular type of device, except as specifically claimed.

(76) The abovementioned cap-free dielectric design described above can be integrated in non-planar transistor devices such as FinFET devices. FIG. **9** is a flow diagram of some embodiments of a method of forming a semiconductor structure, and FIGS. **10A** to **10F** are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments. In some embodiments, a method of forming a semiconductor structure **60** is provided, and the method **50** includes a number of operations (**511**, **512**, **513**, **514**, **515**, **516** and **517**).

(77) FIGS. **10A** to **10F** are schematic drawings illustrating a semiconductor structure at various fabrication stages constructed according to aspects of the present disclosure in one or more embodiments. Referring to FIGS. **9** and **10A**, a substrate **600** including a channel layer is received in operation **511**. In some embodiments of the present disclosure, the substrate **600** may be made of Si or other semiconductor materials. Alternatively or additionally, the substrate **600** may include other elementary semiconductor materials such as Ge. In some embodiments, the substrate **600** is made of a compound semiconductor such as SiC, GaAs, InAs, InP or the like. In some embodiments, the substrate **600** is made of an alloy semiconductor such as SiGe, SiGeC, GaAsP, GaInP, or the like. In some embodiments, the substrate **600** includes an epitaxial layer. For example, the substrate **600** includes an epitaxial layer overlying a bulk semiconductor. In some embodiments, the substrate **600** can include an SOI substrate or a germanium-on-insulator (GOI) substrate. In some embodiments, the substrate **600** can include various doped regions and other suitable features.

(78) In some embodiments, an anti-punch through implant (APT) region (not shown) can be formed in the substrate **600**. The APT region is used to reduce sub-threshold source-to-drain leakage and drain-induced barrier lowering (DIBL). The APT region may be an n-type APT region

or a p-type APT region. In some embodiments, the n-type APT region is obtained by doping with an n-type dopant, such as As, P, or antimony (Sb). In some embodiments, the p-type APT region is obtained by doping with a p-type dopant, such as boron (B), or boron fluorine (BF₂). In some embodiments, a thermal operation can be performed to activate the dopant in the APT region. In some embodiments, the thermal process is performed by a rapid thermal annealing process, but is not limited thereto.

(79) As shown in FIG. 10A, a fin structure **602a** and a fin structure **602b** are disposed over the substrate **600** and protrude from the substrate **600**. Further, each fin structure **602a**, **602b** includes a semiconductor layer. In some embodiments, the fin structure **602a** and the substrate **600** may include the same material, such as Si. Alternatively, the fin structure **602a** and the substrate **600** may include different materials. In some embodiments, the fin structure **602b** may include materials different from those of the substrate **600**. For example, the substrate **600** may include silicon, and the fin structure **602b** may include silicon germanium. The fin structure **602b** may include silicon carbide or Sn, as another example. In some embodiments, the fin structure **602a** and the fin structure **602b** are recognized as including different semiconductor layers. For example but not limited thereto, the fin structure **602a** includes a Si layer while the fin structure **602b** includes a Si_{1-x}Ge_x layer, wherein the germanium content, x, ranges from 0 to 1. In some embodiments, the germanium content may be greater than 0.3. In some embodiments, the Ge content may be greater than 0.35, but the disclosure is not limited thereto.

(80) The fin structures **602a** and **602b** may be formed using a subtractive etch process, by patterning a substrate **600**, and masking the fin structure **602b** while a substance is introduced to alter the material of the fin structure **602b** to include a compound semiconductor material or other material different from the substrate **600** material. The fin structure **602b** may be altered to include the material different from that of the substrate **600** using an epitaxial growth process, implantation process, deposition process, or other type of process. Further, an isolation structure **603** may be disposed between the fin structures **602a** and **602b** to electrically isolate the fin structures **602a** and **602b**.

(81) In some embodiments, the isolation structure **603** may be formed between the fin structures **602a** and **602b**, by depositing insulating material over and between the fin structures **602a** and **602b**, and removing any excess insulating material from over top surfaces of the fin structures **602a** and **602b** using a CMP operation and/or an etch operation, as examples. In other embodiments, the isolation structure **603** may be formed over the substrate **600**, and patterned with a pattern for the fin structures **602a** and **602b**. The fin structures **602a** and/or **602b** are then epitaxially grown within the patterns formed within the isolation structure **603**. Two separate processes may be used to form the fin structure **602a** and the fin structure **602b**, so that they include different materials, depending on the desired functionality of the fin structures **602a** and **602b** for the particular type of FinFET devices they will be used in, for example. In still other embodiments, combinations of subtractive etch processes and epitaxial growth processes may also be used to form the fin structures **602a** and **602b**. The fin structure **602a** may be formed using a subtractive etch process, and at least a portion of the fin structure **602b** may be formed using an epitaxial growth process, for example. The isolation structure **603** may be recessed before or after the formation of the fin structures **602a** and **602b** so that a top surface of the isolation structure **603** is recessed below the top surfaces of the fin structures **602a** and **602b**, as shown in FIG. 10A.

(82) Referring to FIGS. 9 and 10B, a dielectric layer **610** is formed on the fin structures **602a** and **602b** in operation 512. The dielectric layer **610** can include semiconductor oxide. For example, the dielectric layer **610** may include a silicon oxide layer, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **610** may include a first dielectric layer **612a** and a second dielectric layer **612b**, or a first dielectric layer **612a'** and a second dielectric layer **612b**. As shown in FIG. 10B, the first dielectric layer **612a** is sandwiched between the second dielectric layer **612b** and the fin structure **602b**, and the first dielectric layer **612a'** is sandwiched between the second

dielectric layer **612b** and the fin structure **602a**. In some embodiments, the first dielectric layer **612a** and **612a'** may be a III-V compound semiconductor oxide layer. In some embodiments, the III-V compound semiconductor oxide layer may be a native oxide layer that is spontaneously grown on a surface of the fin structures **602a** and **602b**, respectively. For example, the first dielectric layer **612a** may be a native silicon germanium oxide (Si.sub.xGe.sub.yO) layer spontaneously grown on a surface of the SiGe fin structure **602b**, wherein x is between approximately 0.6 and approximately 1, and y is between approximately 0.4 and approximately 0. The first dielectric layer **612a'** may be a native silicon oxide layer spontaneously grown on a surface of the Si fin structure **602a**. In some embodiments, the second dielectric layer **612b** may be a silicon oxide layer. In some embodiments, the second dielectric layer **612b** may be a chemical oxide layer formed by suitable thermal oxidation or deposition. The method of forming the second dielectric layer **612b** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, a thickness of the dielectric layer **610** (i.e., a sum of a thickness of the second dielectric layer **612b** and a thickness of the first dielectric layer **612a** or **612a'**) is between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In other embodiments, the thickness of the dielectric layer **610** is less than 20 Å, but the disclosure is not limited thereto.

(83) Still referring to FIGS. 9 and 10C, a first sacrificial cap **620** including Si is formed on the dielectric layer **610** in operation 513. In some embodiments, the first sacrificial cap **620** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the first sacrificial cap **620** is greater than 20 Å, but the disclosure is not limited thereto. The method of forming the first sacrificial cap **620** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, the first sacrificial cap **620** is conformally formed to cover the fin structures **602a** and **602b**, the substrate **600** and the isolation structure **503**, but the disclosure is not limited thereto. In some embodiments, the thickness of the first sacrificial cap **620** is sufficiently thick so that the space between the fin structures **602a** and **602b** is filled by the first sacrificial cap **620**, but the disclosure is not limited thereto.

(84) Referring to FIGS. 9 and 10D, the substrate **600** is annealed to transform the first sacrificial cap **620** into a second sacrificial cap **620'** in operation 614. In some embodiments, an anneal **630** is performed, wherein the details of the anneal **630** can be similar to those of anneals described above; therefore, similar details are omitted in the interest of brevity. During the anneal **630**, Ge atoms diffuse from the fin structure **602b** and from the first dielectric layer **612a** over the fin structure **602b**. Further, Ge atoms may diffuse upwardly and pass the second dielectric layer **612b**, and stay with Si atoms in the first sacrificial cap **620**. Accordingly, the first sacrificial cap **620** including silicon is transformed into the second sacrificial cap **620'** including silicon germanium. In other words, the second sacrificial cap **620'** may include Ge atoms from the fin structure **602b** and from the first dielectric layer **612a**. It should be noted that the thickness of the first sacrificial cap **620** (now the second sacrificial cap **620'**) is greater than 20 Å in order to provide sufficient accommodations for the Ge atoms.

(85) In some embodiments, the first dielectric layer **612a'** including silicon oxide and the second dielectric layer **612b** including silicon oxide over the fin structure **602a** are densified during the anneal **630**, and thus quality of the silicon oxide dielectric layer **610'** over the fin structure **602a** is improved.

(86) As described above, Ge atoms may diffuse from the semiconductor layer in the fin structure **602b**. Accordingly, a portion of the fin structure **602b** may lose Ge atom. In some embodiments, it is observed that a portion of the fin structure **602b**, usually an outer portion, is likely to exhibit Ge diffusion. Accordingly, the portion having a lower Ge concentration is recognized and defined as a first portion **602O**, while the other portion, usually a portion located inside of the first portion **602O**, having the Ge concentration greater than that of the first portion **602O**, is recognized and defined as a second portion **602I**. In some embodiments, the Ge concentration in the second portion

602I may be substantially equal to an original Ge concentration in the fin structure **602b**. The first portion **602O** of the fin structure **602b** has a first Ge concentration prior to the annealing of the substrate **600** and a second Ge concentration after the annealing of the substrate **600**. In some embodiments, the first Ge concentration in the first portion **602O** is substantially equal to the Ge concentration in the second portion **602I**, which is the original Ge concentration. The second Ge concentration in the first portion **602O** is less than the first Ge concentration in the first portion **602O** and the Ge concentration in the second portion **602I**. For example but not limited thereto, the first Ge concentration in the first portion **602O** and the Ge concentration in the second portion **602I** may be greater than approximately 30%, while the second Ge concentration in the first portion **602O** may be less than approximately 25%. Accordingly, a Ge concentration in an interface **640** between the dielectric layer **610** and the fin structure **602b** is reduced from greater than approximately 30% to less than approximately 25%, but the disclosure is not limited thereto.

(87) Further, Ge atoms in the first dielectric layer **612a** may also diffuse into the first sacrificial cap **620**, and thus the first dielectric layer **612a**, which includes a silicon germanium oxide layer may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **610'** including silicon oxide may be formed after operation **314**. In some embodiments, the dielectric layer **610'** includes the original second dielectric layer **612b** including silicon oxide and the first dielectric layer **612a** previously including silicon germanium oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **610'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **610'** is less than 1.5%.

(88) Referring to FIGS. **9** and **10E**, the second sacrificial cap **620'** is removed to expose the dielectric layer **610'** in operation **515**. Referring to FIGS. **9** and **10F**, a semiconductor gate layer **650** is formed over the dielectric layer **610'** in operation **516**. In some embodiments, the dielectric layer **610'** serves as a gate dielectric layer for a transistor device, and a thickness of the dielectric layer **610'** may be between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In such embodiments, the gate layer can include a semiconductor material; for example, a doped polysilicon is formed to serve as the semiconductor gate layer **650**.

(89) Still referring to FIG. **10F**, in some embodiments, a patterned hard mask **652** is formed on the semiconductor gate layer **650**, and the semiconductor gate layer **650** is patterned through the patterned hard mask, such that a gate structure including the semiconductor gate layer **650** and the dielectric layer **610'** is obtained. Accordingly, a channel layer is formed in the fin structures **602a** and **602b** beneath the gate structure.

(90) Referring to FIG. **11**, in some embodiments, source/drain extension regions (not shown) for different devices (i.e., nMOS device and pMOS device) can be formed in the substrate **600** at two sides of the gate structure and spacer (not shown) can be formed on sidewalls of the gate structure. A source/drain **664a** and a source/drain **664b** for different devices can be formed in the substrate **600** at two sides of the gate structure and the spacers in operation **517**. In some embodiments, at least the source/drain **664b** can be a strained source/drain (S/D) structure for different devices. The strained S/D structures can be formed by growing a strained material in recesses (not shown) by an epitaxial (epi) operation. In some embodiments, a lattice constant of the strained material may be different from a lattice constant of the substrate. In some embodiments, materials for forming the strained S/D structures can be similar to materials described above; therefore, similar details are omitted for brevity. Accordingly, a semiconductor structure such as an non-planar MOSFET device (i.e., a FinFET device) **60** is obtained, as shown in FIG. **11**.

(91) The method of forming the semiconductor structure **50** uses a cap-free dielectric design. According to the method **50**, the Ge concentration in the dielectric layer **610'** is less than 3% or even less than 1.5%. Consequently, gate leakage current (J_g) is reduced by the dielectric layer **610'** and reliability of the MOSFET device **60** including the SiGe fin structure **602b** is improved. Further, interface trap density (D_{it}) and density of interface states are also reduced by the method **50**. Therefore the carrier mobility is improved by the SiGe channel layer in the fin structure **602b**.

while the carrier mobility degradation issue caused by high Dit and density of interface states is mitigated.

(92) As described above, in some embodiments, metal gate approach can be adopted to use a high-k gate dielectric and a metal gate electrode to improve device performance. Therefore, the method **50** can further include a number of operations (**521**, **522**, **523**, **524**, **525**, **526**, **527** and **528**). In some embodiments, operation **521** can be performed after operation **517**, but the disclosure is not limited thereto. In some embodiments, operations **527** and **528** can be performed after operation **521**, but the disclosure is not limited thereto. In other embodiments, operations **522**, **523**, **524**, **525**, **526**, **527** and **528** can be performed after operation **521**, but the disclosure is not limited thereto.

(93) Referring to FIG. **12**, as described above, devices in an I/O region and a core region are expected to work under different operation voltages, and thus different operations can be performed to form the devices in the I/O region and the core region. For example, in some embodiments, the above-mentioned operations **511**, **512**, **513**, **514**, **515**, **516** and **517** are performed to simultaneously form a FinFET device **60a** including a polysilicon gate electrode **650** in an I/O region **604a** and a FinFET device **60b** including polysilicon gate electrode **650** in a core region **604b**, as shown in FIG. **12**.

(94) In some embodiments, a dielectric structure (not shown) can be formed over the substrate **600**, such that the FinFET device **60a** and the FinFET device **60b** are both embedded in the dielectric structure. In some embodiments, the dielectric structure can include a CESL (not shown) and various dielectric layers (e.g., an inter-layer dielectric (ILD) layer) formed over the substrate after the forming of the strained S/D structures. The materials used to form the CESL and the ILD layers can be similar to materials described above; therefore, similar details are omitted for brevity.

(95) In some embodiments, the operations **521**, **527** and **528** can be performed to form a FinFET device **60a** having a metal gate electrode in the I/O region **604a**, but the disclosure is not limited thereto. Accordingly, only the FinFET device **60a** in the I/O region **604a** is shown in FIGS. **13A** to **13C**. In some embodiments, the semiconductor gate layer **650** (i.e., the polysilicon gate electrode) serves as a sacrificial gate structure, also known as a dummy gate structure, in a replacement gate (RPG) approach. Referring to FIGS. **9** and **13A**, in some embodiments, after the dielectric structure is formed, a planarization process, such as a CMP operation, may be performed to remove a portion of the dielectric structure, a portion of the spacer and the patterned hard mask **652**. Consequently, a top surface of the semiconductor gate structure **650** is exposed. Subsequently, the semiconductor gate structure **650** is removed in operation **521**. Consequently, a gate trench **654** is formed within the dielectric structure and the spacers. Further, the first dielectric layer **610'** is exposed through the gate trench **654**, as shown in FIG. **13A**.

(96) Referring to FIGS. **9** and **13B**, a high-k dielectric layer **680** is formed on the first dielectric layer **610'** in operation **527**. The materials used to form the high-k dielectric layer **680** can be the same as materials described above; therefore, those details are omitted for brevity. In such embodiments, the dielectric layer **610'** serves as an interfacial layer (IL) between the high-k dielectric layer **680** and the fin structures **602a** and **602b**.

(97) Referring to FIGS. **9** and **13C**, metal gate structures **682a** and **682b** are formed over the high-k dielectric layer **680** in operation **528**. In some embodiments, the metal gate structure **682a** can include at least a barrier metal layer (not shown), a work function metal layer **684a** for the nMOS device and a gap-filling metal layer **686**, while the metal gate structure **682b** can include at least a barrier metal layer (not shown), a work function metal layer **684b** for a pMOS device and a gap-filling metal layer **686**. Materials for forming the barrier layer, the work function metal layers **684a** and **684b**, and the gap-filling layer **686** can be similar to materials described above; therefore, similar details are omitted for brevity.

(98) Accordingly, a FinFET device **60a** with metal gate is obtained in the I/O region **604a**, according to the method **50**. The FinFET device can be an n-type FinFET device and/or a p-type FinFET device. The n-type and p-type FinFET devices can be electrically connected to form a

CMOS device in the I/O region **604a**, as shown FIG. **13C**.

(99) In contrast to the device in the I/O region **604a**, the FinFET device **60b** having a metal gate electrode in the core region **604b** can be formed by the operations **521**, **522**, **523**, **524**, **525**, **526**, **527** and **528**, but the disclosure is not limited thereto. Accordingly, only the device **60b** in the core region **604b** is shown in FIGS. **14A** to **14H**. In some embodiments, the semiconductor gate layer **650** (i.e., the polysilicon gate electrode) serves as a sacrificial gate, also known as a dummy gate, in a replacement gate (RPG) approach. Referring to FIGS. **9** and **14A**, as described above, after the CESL and the ILD layer are deposited, a portion of the dielectric structure, a portion of the spacer and the patterned hard mask are removed to expose a top surface of the semiconductor gate layer **650**, and the semiconductor gate structure **650** is removed in operation **321**. Consequently, a gate trench **654** is formed within the dielectric layer and the spacers. Further, the dielectric layer **610'** is exposed through the gate trench **654**.

(100) Referring to FIGS. **9** and **14B**, the dielectric layer **610'** is removed in operation **522**.

Consequently, the fin structures **602a** and **602b** in the core region **604b** are exposed through the gate trench **654**.

(101) Referring to FIGS. **9** and **14C**, another dielectric layer **614** is formed on the fin structures **602a** and **602b** in operation **523**. The dielectric layer **614** can include semiconductor oxide. For example, the dielectric layer **614** may include a silicon oxide layer, but the disclosure is not limited thereto. In some embodiments, the dielectric layer **614** may include a first dielectric layer **616a** and a second dielectric layer **616b** over the fin structure **602b**, or include a first dielectric layer **616a'** and a second dielectric layer **616b** over the fin structure **602a**. As shown in FIG. **14C**, the first dielectric layer **616a** is sandwiched between the second dielectric layer **616b** and the fin structure **602b** while the first dielectric layer **616a'** is sandwiched between the second dielectric layer **616b** and the fin structure **602a**. In some embodiments, the first dielectric layer **616a** may be a III-V compound semiconductor oxide layer. In some embodiments, the III-V compound semiconductor oxide layer may be a native oxide layer that is spontaneously grown on a surface of the fin structures **602a** and **602b**. For example, the first dielectric layer **616a** may be a native silicon germanium oxide layer spontaneously grown on a surface of the fin structure **602b**. In some embodiments, the first dielectric layer **616a'** may be a native silicon oxide layer spontaneously grown on a surface of the fin structure **602a**. In some embodiments, the second dielectric layer **616b** can be a silicon oxide layer. In some embodiments, the second dielectric layer **616b** may be a chemical oxide layer formed by suitable thermal oxidation or deposition. The method for forming the second dielectric layer **616b** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, a thickness of the dielectric layer **614** is less than approximately 20 Å, but the disclosure is not limited thereto. In some embodiments, when the second dielectric layer **616b** of the dielectric layer **614** is formed by deposition, the second dielectric layer **616b** may cover top surfaces and sidewalls of the fin structures **602a** and **602b**, and top surfaces of the isolation structure **603**, as shown in FIG. **14C**, but the disclosure is not limited thereto.

(102) Referring to FIGS. **9** and **14D**, a third sacrificial cap **622** including Si is formed on the dielectric layer **614** in operation **524**. In some embodiments, the third sacrificial cap **622** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the third sacrificial cap **622** is greater than 10 Å, but the disclosure is not limited thereto. The method of forming the third sacrificial cap **622** can be similar to methods described above; therefore, those details are omitted in the interest of brevity.

(103) Referring to FIGS. **9** and **14E**, the substrate **600** is annealed to transform the third sacrificial cap **622** into a fourth sacrificial cap **622'** in operation **525**. In some embodiments, an anneal **632** is performed, wherein the details of the anneal **632** can be similar to those of anneals described above; therefore, similar details are omitted in the interest of brevity. During the anneal **632**, Ge atoms may diffuse from the fin structure **602b** and from the first dielectric layer **616a**. Further, Ge

atoms may diffuse upwardly and pass the second dielectric layer **616b**, and stay with Si atoms in the third sacrificial cap **622**. Accordingly, the third sacrificial cap **622** including silicon is transformed into the fourth sacrificial cap **622'** including silicon germanium. In other words, the fourth sacrificial cap **622'** may include Ge atoms from the fin structure **602b** and from the first dielectric layer **616a**. It should be noted that the thickness of the third sacrificial cap **622** (now the fourth sacrificial cap **622'**) is greater than 10 Å in order to provide sufficient accommodations for the Ge atoms.

(104) It should be noted that the first portion **602O'** of the fin structure **602b** in the core region **604b** may undergo the anneal twice (i.e., the anneal **630** and the anneal **632**); therefore, more Ge atoms may diffuse from the first portion **602O'** of the fin structure **602b** in the core region **604b**. Accordingly, the Ge concentration in the first portion **602O'** of the fin structure **602b** in the core region **604b** may be further reduced.

(105) Further, Ge atoms in the first dielectric layer **616a** may also diffuse into the third sacrificial cap **622**, and thus the first dielectric layer **616a**, which includes a silicon germanium oxide layer, may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **614'** including silicon oxide may be formed after the operation **525**. In some embodiments, the dielectric layer **614'** includes the original second dielectric layer **616b** including silicon oxide and the first dielectric layer **616a** previously including silicon germanium oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **614'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **614'** is less than 1.5%.

(106) Referring to FIGS. **9** and **14F**, the fourth sacrificial cap **622'** is removed to expose the dielectric layer **614'** in operation **526**. In some embodiments, the dielectric layer **614'** serves as an IL for a transistor device, and a thickness of the dielectric layer **614'** may be less than approximately 20 Å, but the disclosure is not limited thereto.

(107) Referring to FIGS. **9** and **14G**, a high-k dielectric layer **680** is formed on the dielectric layer **614'** in operation **527**. In some embodiments, materials used to form the high-k dielectric layer **680** can be similar to materials described above; therefore, similar details are omitted for brevity.

(108) Referring to FIGS. **9** and **14H**, metal gate structures **682a** and **682b** are formed over the high-k dielectric layer **680** in operation **528**. In some embodiments, the metal gate structure **682a** can include at least a barrier metal layer (not shown), a work function metal layer **684a** for the nMOS device and a gap-filling metal layer **686**, while the metal gate structure **682b** can include at least a barrier metal layer (not shown), a work function metal layer **684b** for a pMOS device and a gap-filling metal layer **686**. Materials for forming the barrier layer, the work function metal layers **684a** and **684b**, and the gap-filling layer **686** can be similar to materials described above; therefore, similar details are omitted for brevity.

(109) Accordingly, FinFET devices with metal gates are obtained in the core region **604b**, according to the method **50**. The FinFET device can be an n-type FinFET device or a p-type FinFET device. The n-type and p-type FinFET devices can be electrically connected to form a CMOS device in the core region **604b**, as shown FIG. **14H**.

(110) According to the method **50**, when the FinFET device including metal gate **60a** is required in the I/O region **604a**, operations **527** and **528** can be performed after operation **521**, and when the FinFET device including metal gate **60b** is required in the core region **604b**, operations **522** to operation **528** can be performed after operation **521**. In some embodiments, the semiconductor gate layer **650** in both of the device **60a** and **60b** can be removed simultaneously, in operation **521**, and a protecting layer (not shown) can be formed in the I/O region **604a** after the removal of the semiconductor gate layer in order to protect the dielectric layer **610'** in the device **60a** in the I/O region **604a**. The operations **522** to **526** can be performed on the device **60b** in the core region **604b** without impacting the device **60a** in the I/O region **604a**. In such embodiments, the protecting layer can be removed after the removal of the fourth sacrificial cap **622'** (i.e., operation **526**), and the high-k gate dielectric layer **680** and the metal gate layers **682a** and **682b** can be formed in the gate

trenches **654** in the SiGe MOSFET device **60a** in the I/O region **604a** and in the MOSFET device **60b** in the core region **604b**.

(111) Referring to FIG. 15, accordingly, a semiconductor structure is provided. The semiconductor structure includes a substrate **600** including a first region **604a** (i.e., an I/O region) and a second region **604b** (i.e., a core region). The semiconductor structure may include fin structures **602a** including silicon channel layers disposed in the first region **604a** and the second region **604b**, and fin structures **602b** including silicon germanium channel layers disposed in the first region **604a** and the second region **604b**. In some embodiments, the fin structure **602b** in the first region **604a** includes a first portion **602O** and a second portion **602I**, wherein the Ge concentration in the first portion **602O** is less than that in the second portion **602I**. In some embodiments, the fin structure **602b** in the second region **604b** includes a first portion **602O'** and the second portion **602I**, wherein the Ge concentration in the first portion **602O'** is less than that in the second portion **602I**. In some embodiments, the second portions **602I** of the fin structures **602b** in both of the first and second regions **604a** and **604b** may include the same Ge concentration. However, because the first portion **602O'** of the fin structure **602b** in the core region **604b** may undergo the anneal twice (i.e., the anneal **630** and the anneal **632**), more Ge atoms may diffuse from the first portion **602O'** of the fin structure **602b** in the core region **604b**. Accordingly, the Ge concentration in the first portion **602O'** of the fin structure **602b** in the core region **604b** may be less than the Ge concentration in the first portion **602O** of the fin structure **602a** in the I/O region **604a**.

(112) Still referring to FIG. 15, the semiconductor structure further includes a dielectric layer **610'** having a thickness **T1** disposed on the fin structures **602a** and **602b** in the first region **604a** and a dielectric layer **614'** having a thickness **T2** disposed on the fin structures **602a** and **602b** in the second region **604b**. In some embodiments, the dielectric layer **610'** and the dielectric layer **614'** may serve as an IL, and a high-k dielectric layer **680** is disposed on the dielectric layer **610'** and the dielectric layer **614'**, respectively, as shown in FIG. 15. In such embodiments, a metal gate electrode can be formed on the high-k dielectric layer **680**, as shown in FIG. 15. The metal gate electrode may include different work function metal layers for n-type or p-type FinFET devices. In some embodiments, different p-type work function metal layers can be formed in the I/O region **604a** and the core region **604b**. Similarly, different n-type work function metal layers can be formed in the I/O region **604a** and the core region **604b**.

(113) In some embodiments, the thickness **T1** of the dielectric layer **610'** in the first region **604a** is greater than the thickness **T2** of the dielectric layer **614'** in the second region **604b**. In some embodiments, the thickness **T1** of the dielectric layer **610'** in the first region **604a** is between approximately 10 Å and approximately 50 Å, but the disclosure is not limited thereto. In some embodiments, the thickness **T2** of the dielectric layer **614'** in the second region **604b** is less than 20 Å, but the disclosure is not limited thereto. Additionally, the dielectric layer **610'** is in contact with the fin structures **602a** and **602b** and the isolation structure **603** in the I/O region **604a**, and the dielectric layer **614'** is in contact with the fin structures **602a** and **602b** and the isolation structure **603** in the core region **604b**. In some embodiments, a thickness of the portion of the dielectric layer **610'** that is in contact with the isolation structure **603** is less than a thickness of the portion of the dielectric layer **610'** that are in contact with the fin structures **602a** and **602b** in the first region **604a**. Similarly, a thickness of the portion of the dielectric layer **614'** that is in contact with the isolation structure **603** is less than a thickness of the portions of the dielectric layer **614'** that are in contact with the fin structure **602a** and **602b** in the second region **604b**.

(114) Still referring to FIG. 15, the semiconductor structure further includes a gate electrode disposed on the high-dielectric layers **680** in the first region **604a** and the second region **604b**, respectively. The gate electrode can be a metal gate layer **682a** or **682b**, but the disclosure is not limited thereto.

(115) According to the method **50**, the Ge concentrations in the dielectric layers **610'** and **614'** are less than 3% or even less than 1.5%, while the Ge concentration in the dielectric layer **614'** may be

less than that in the dielectric layer **610'**. Consequently, gate leakage current (J_g) is reduced by the dielectric layers **610'** and **614'** and reliability of the FinFET devices **60a** and **60b** in both of the first region **604a** and second region **604b** is improved. Further, interface trap density (D_{it}) and density of interface states are also reduced by the method **50**. Therefore the carrier mobility of the FinFET devices **60a** and **60b** is improved by the SiGe channel layer **602** while the carrier mobility degradation issue caused by high D_{it} and density of interface states is mitigated.

(116) In a FinFET device, a gate electrode is adjacent to three side surfaces of a fin structure with a gate dielectric layer interposed therebetween. Because the gate structure surrounds the fin structure on three surfaces, the transistor essentially has three gates controlling the current through the fin or channel region. However, the fourth side, which is the bottom part of the channel region, is far away from gate electrode and is not under close gate control. Unlike in a FinFET device, in a GAA FET device all side surfaces of the channel region are surrounded by the gate electrode, which allows for fuller depletion in the channel region and results in fewer short-channel effects due to a steeper sub-threshold current swing (SS) and smaller drain-induced barrier lowering (DIBL).

(117) The cap-free dielectric design described above can be integrated in non-planar transistor devices such as GAA FET devices. FIGS. **16**, **17A** to **17F** and **18** are schematic drawings illustrating the method of forming the semiconductor structure at various fabrication stages.

(118) Referring to FIG. **16**, a substrate **700** is received. In some embodiments, the substrate **700** may be a semiconductor substrate such as a Si substrate. A dummy gate (not shown), a spacer **702**, a plurality of nanowires **704** and source/drain regions **706** may be formed over the substrate **700**. Further, a dielectric structure **710** is disposed over the substrate **700**. In some embodiments, the dielectric structure **710** can include a CESL **712** and various dielectric layers (e.g., an inter-layer dielectric (ILD) layer) **714** formed over the substrate **700** after the forming of the source/drain regions **706**. Materials for forming the CESL **712** and the ILD layer **714** can be similar to materials described above; therefore, similar details are omitted for brevity.

(119) In some embodiments, the source/drain regions **706** may be formed by performing an epitaxial growth operation that provides an epitaxy material. In various embodiments, the source/drain regions **706** may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. In some embodiments, the source/drain regions **706** may be in-situ doped during the epi process. For example, in some embodiments, the source/drain regions **706** may be doped with boron. In some embodiments, the source/drain regions **706** may be doped with carbon to form a Si:C source/drain, phosphorus to form a Si:P source/drain, or both carbon and phosphorous to form SiCP source/drain. In some embodiments, the source/drain regions **706** are not in-situ doped, and an implantation process is instead performed to dope the source/drain regions **706**.

(120) Referring to FIG. **16**, the dummy gate is removed to form a gate trench **716** in the dielectric structure **710** and the spacer **702**, and the nanowires **704** are exposed through the gate trench **716**.

(121) Referring to FIGS. **17A** to **17F**, which are cross-sectional views taken along a line I-I' of FIG. **16**, the nanowires **704** are exposed through the gate trench **718**. In some embodiments, each nanowire includes the semiconductor layer **704** serving as a channel region. In other words, the nanowire **704** includes the channel layer. In some embodiments, the nanowires **704** can be slightly etched to obtain various desirable dimensions and shapes, and the various desired dimensions and shapes may be chosen based on device performance considerations.

(122) Referring to FIG. **17B**, a dielectric layer **730** is formed to surround or wrap each nanowire **704**. In some embodiments, the dielectric layer **730** may include a first dielectric layer **732a** and a second dielectric layer **732b**. The dielectric layer **730** (including the first dielectric layer **732a** and the second dielectric layer **732b**) can be similar to the aforementioned dielectric layer **610**, therefore those details are omitted for brevity. Because the second dielectric layer **732b** is formed by deposition, the second dielectric layer **732b** may cover sidewalls and bottom of the gate trench **716**, though not shown. In other embodiments, the thickness of the dielectric layer **730** is less than 20 Å, but the disclosure is not limited thereto.

(123) Referring to FIG. 17C, a first sacrificial cap **740** including Si is formed on the dielectric layer **730**. In some embodiments, the first sacrificial cap **740** may include an amorphous silicon layer or a polysilicon layer. In some embodiments, a thickness of the first sacrificial cap **740** is greater than 20 Å, but the disclosure is not limited thereto. The method of forming the first sacrificial cap **740** can be similar to methods described above; therefore, similar details are omitted in the interest of brevity. In some embodiments, the first sacrificial cap **740** wraps around and surrounds each nanowire **704** and the dielectric layer **730**. In some embodiments, space between the nanowires **704** is filled by the first sacrificial cap **740** when the distance between adjacent nanowire **704** is less than 20 Å, as shown in FIG. 17C, but the disclosure is not limited thereto.

(124) Referring to FIG. 17D, the substrate **700** is annealed to transform the first sacrificial cap **740** into a second sacrificial cap **740'** including silicon germanium. In some embodiments, an anneal **750** is performed, wherein the details of the anneal **750** can be similar to those of anneals described above; therefore, similar details are omitted in the interest of brevity. During the anneal **750**, Ge atoms may diffuse from the nanowires **704** and from the first dielectric layer **732a**. Further, Ge atoms may diffuse upwardly and pass the second dielectric layer **732b**, and stay with Si atoms in the first sacrificial cap **740**. Accordingly, the first sacrificial cap **740** including Si is transformed into the second sacrificial cap **740'** including SiGe. In other words, the second sacrificial cap **740'** may include Ge atoms from the nanowires **704** and from the first dielectric layer **732a**. It should be noted that the thickness of the first sacrificial cap **740** (now the second sacrificial cap **740'**) is greater than 20 Å in order to provide sufficient accommodations for the Ge atoms.

(125) As described above, Ge atoms may diffuse from the nanowires **704**. Accordingly, a portion of each nanowire **704** may lose Ge atoms. In some embodiments, it is observed that a portion of each nanowire **704**, usually an outer portion, is likely to exhibit Ge diffusion. Accordingly, the portion having lower Ge concentration is recognized and defined as a first portion **704O** while the other portion, usually a portion inside of the first portion **704O**, having the Ge concentration greater than that of the first portion **704O**, is recognized and defined as a second portion **704I**. In some embodiments, the Ge concentration in the second portion **704I** may be substantially equal to an original Ge concentration in the nanowires **704**. The first portion **704O** of each nanowire **704** has a first Ge concentration prior to the annealing of the substrate **700** and a second Ge concentration after the annealing of the substrate **700**. In some embodiments, the first Ge concentration in the first portion **704O** is substantially equal to the Ge concentration in the second portion **704I**, which is the original Ge concentration. The second Ge concentration in the first portion **704O** is lower than the first Ge concentration in the first portion **704O** and the Ge concentration in the second portion **704I**. For example but not limited thereto, the first Ge concentration in the first portion **704O** and the Ge concentration in the second portion **704I** may be greater than approximately 30%, while the second Ge concentration in the first portion **704O** may be less than approximately 25%. Accordingly, a Ge concentration in an interface **742** between the dielectric layer **730** and each nanowire **704** is reduced from greater than approximately 30% to less than approximately 25%, but the disclosure is not limited thereto.

(126) Further, Ge atoms in the first dielectric layer **732a** may also diffuse into the first sacrificial cap **740**, and thus the first dielectric layer **732a** including silicon germanium oxide layer may be transformed into a silicon oxide layer. Accordingly, a dielectric layer **730'** including silicon oxide may be formed. In some embodiments, the dielectric layer **730'** includes the original second dielectric layer **732b** including silicon oxide and the first dielectric layer **732a** previously including silicon oxide and now including silicon oxide. In some embodiments, the Ge concentration in the dielectric layer **730'** is less than 3%. In some embodiments, the Ge concentration in the dielectric layer **730'** is less than 1.5%.

(127) Referring to FIG. 17E, the second sacrificial cap **740'** is removed to expose the dielectric layer **730'**. In some embodiments, the dielectric layer **730'** serves as an IL for a transistor device.

(128) Referring to FIG. 17F, a high-k dielectric layer **760** is formed on the dielectric layer **730'**.

Materials used to form the high-k dielectric layer **760** can be similar to materials described above; therefore, similar details are omitted for brevity.

(129) Referring to FIG. **17F**, a metal gate structure **762** is formed over the high-k dielectric layer **760**. In some embodiments, the metal gate structure **762** can include at least a barrier metal layer (not shown), a work function metal layer **764** and a gap-filling metal layer **766**. Materials for forming the barrier layer, the work function metal layer **764** and the gap-filling layer **766** can be similar to materials described above; therefore, similar details are omitted for brevity.

(130) Accordingly, a GAA FET device **70** with metal gate is obtained. The GAA FET device can be an n-type GAA FET device or a p-type GAA FET device. Referring to FIGS. **17F** and **18**, accordingly, a semiconductor structure such as the GAA FET device **70** is provided. The semiconductor structure includes a substrate **700** and a plurality of nanowires **704** suspended over the substrate **700**. Each nanowire **704** includes the semiconductor layer where a channel region is to be formed. In some embodiments, each nanowire **704** includes a channel layer. Each nanowire **704** (i.e., the semiconductor layer, or the channel layer) includes Ge compound such as SiGe. In some embodiments, each nanowire **704** includes a first portion **704O** and a second portion **704I**, wherein the Ge concentration in the first portion **704O** is less than that in the second portion **704I**.

(131) Still referring to FIGS. **17F** and **18**, the semiconductor structure **70** further includes a dielectric layer **730'** serving as an IL, a high-k dielectric layer **760** disposed on the dielectric layer **730'** and a metal gate layer **762** disposed on the high-k dielectric layer **760**. Further, the metal gate layer **762** fills the gate trench **718**, as shown in FIG. **18**.

(132) The Ge concentration in the dielectric layer **730'** is less than 3% or even less than 1.5%. Consequently, gate leakage current (J_g) is reduced by the dielectric layer **730'** and reliability of the GAA FET device **70** is improved. Further, interface trap density (D_{it}) and density of interface states are also reduced. Therefore the carrier mobility is improved by the silicon germanium channel layer in each nanowire **704** while carrier mobility degradation issue caused by high D_{it} and density of interface states is mitigated.

(133) Accordingly, the present disclosure provides a cap-free design for the gate dielectric layer on a silicon germanium channel layer. According to the provided cap-free design, a dielectric layer is formed over a silicon germanium layer where a channel is to be formed, and a sacrificial semiconductor layer is formed on the dielectric layer. An anneal is subsequently performed. During the anneal, Ge atoms may diffuse from the silicon germanium layer, pass the first dielectric layer, and remain in the semiconductor sacrificial cap. The semiconductor sacrificial cap including the Ge atoms is then removed. Further, an interface between the silicon germanium layer and the dielectric layer can be improved during the anneal. Consequently, a dielectric layer having low interface trap density is obtained without SiGe loss. The dielectric layer used in a pMOS device including the silicon germanium layer where the channel is to be formed reduces gate leakage current and thus improves reliability. Accordingly, the cap-free design for the gate dielectric layer on a silicon germanium channel layer offers an opportunity to achieve high mobility and improve transistor performance.

(134) According to one embodiment of the present disclosure, a semiconductor structure is provided. The semiconductor structure includes a first fin structure and a second fin structure, a first dielectric layer disposed over the first fin structure, a second dielectric layer disposed over the second fin structure, a first gate electrode disposed over the first dielectric layer, and a second gate electrode disposed over the second dielectric layer. A thickness of the first dielectric layer and a thickness of the second dielectric layer are equal. The second fin structure includes an outer region and an inner region, and a Ge concentration in the outer portion is less than Ge concentration in the inner portion.

(135) According to one embodiment of the present disclosure, a semiconductor structure is provided. The semiconductor structure includes a first fin structure, a second fin structure, a first semiconductor oxide layer disposed over the first fin structure, a second semiconductor oxide layer

disposed over the second fin structure, a first gate electrode disposed over the first dielectric layer, and a second gate electrode disposed over the second dielectric layer. The first semiconductor oxide layer includes a first semiconductor material. The second semiconductor oxide layer includes the first semiconductor material and a second semiconductor material. The second fin structure includes the first semiconductor material and the second semiconductor material, and a concentration of the second semiconductor material is outwardly reduced. A thickness of the first semiconductor oxide layer and a thickness of the second semiconductor oxide layer are equal.

(136) According to one embodiment of the present disclosure, a semiconductor structure is provided. The semiconductor structure includes a Si fin structure, a SiGe fin structure, a high-k gate dielectric layer over the Si fin structure and the SiGe fin structure, a first dielectric layer between the high-k gate dielectric layer and the Si fin structure, a second dielectric layer between the high-k gate dielectric layer and the SiGe fin structure, a first metal gate electrode over the high-k gate dielectric layer, and a second metal gate electrode over the high-k gate dielectric layer. The first dielectric layer includes silicon oxide and the second dielectric layer comprises silicon germanium oxide. A thickness of the first dielectric layer and a thickness of the second dielectric layer are equal.

(137) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure comprising: a first fin structure and a second fin structure; a first dielectric layer disposed over the first fin structure, and a second dielectric layer disposed over the second fin structure, wherein a thickness of the first dielectric layer and a thickness of the second dielectric layer are equal; and a first gate electrode disposed over the first dielectric layer, and a second gate electrode disposed over the second dielectric layer, wherein the second fin structure includes an outer portion and an inner portion, and a germanium (Ge) concentration in the outer portion is less than a Ge concentration in the inner portion, wherein the second dielectric layer comprises Ge, and a Ge concentration in the second dielectric layer is less than 3%.
2. The semiconductor structure of claim 1, wherein a Ge concentration in an interface between the second dielectric layer and the second fin structure is less than approximately 25%.
3. The semiconductor structure of claim 1, wherein the Ge concentration in the outer portion of the second fin structure is less than approximately 25%.
4. The semiconductor structure of claim 1, wherein the first gate electrode and the second electrode comprise different metal materials.
5. The semiconductor structure of claim 1, further comprising: a first high-k dielectric layer between the first gate electrode and the first dielectric layer; and a second high-k dielectric layer between the second gate electrode and the second dielectric layer.
6. The semiconductor structure of claim 5, wherein a thickness of the first high-k dielectric layer and a thickness of the second high-k dielectric layer are equal.
7. The semiconductor structure of claim 5, wherein the first high-k dielectric layer and the second high-k dielectric layer are coupled to each other.
8. The semiconductor structure of claim 1, wherein the first dielectric layer and the second dielectric layer are coupled to each other.

9. A semiconductor structure comprising: a first fin structure and a second fin structure; a first semiconductor oxide layer disposed over the first fin structure, wherein the first semiconductor oxide layer comprises a first semiconductor material; a second semiconductor oxide layer disposed over the second fin structure, wherein the second semiconductor oxide layer comprises a first semiconductor material and a second semiconductor material; and a first gate electrode disposed over the first dielectric layer, and a second gate electrode disposed over the second dielectric layer; wherein the second fin structure includes the first semiconductor material and the second semiconductor material, and a concentration of the second semiconductor material is outwardly reduced, and wherein a thickness of the first semiconductor oxide layer and a thickness of the second semiconductor oxide layer are equal.
10. The semiconductor structure of claim 9, wherein the first semiconductor material is silicon, and the second semiconductor material is germanium.
11. The semiconductor structure of claim 9, wherein a concentration of the second semiconductor material in the second dielectric layer is less than approximately 3%.
12. The semiconductor structure of claim 9, wherein the first gate electrode and the second electrode comprise different metal materials.
13. The semiconductor structure of claim 9, further comprising: a first high-k dielectric layer between the first gate electrode and the first dielectric layer; and a second high-k dielectric layer between the second gate electrode and the second dielectric layer.
14. A semiconductor structure comprising: a silicon (Si) fin structure and a silicon germanium (SiGe) fin structure; a high-k dielectric layer over the Si fin structure and the SiGe fin structure; a first dielectric layer between the high-k dielectric layer and Si fin structure, and a second dielectric layer between the high-k dielectric layer and the SiGe fin structure; and a first metal gate electrode over the high-k dielectric layer and a second metal gate electrode over the high-k dielectric layer, wherein the first dielectric layer comprises silicon oxide, the second dielectric layer comprises silicon germanium oxide, and a thickness of the first dielectric layer and a thickness of the second dielectric layer are equal.
15. The semiconductor structure of claim 14, wherein the SiGe fin structure further comprises an outer portion and an inner portion inside the outer portion, and a Ge concentration in the outer portion is less than a Ge concentration in the inner portion.
16. The semiconductor structure of claim 15, wherein the Ge concentration in the outer portion is less than approximately 25%.
17. The semiconductor structure of claim 15, wherein the Ge concentration in the inner portion is greater than approximately 30%.
18. The semiconductor structure of claim 14, wherein a Ge concentration in the second dielectric layer is less than approximately 3%.
19. The semiconductor structure of claim 14, wherein the thickness of the first dielectric layer and the thickness of the second dielectric layer are between approximately 10 angstroms and approximately 50 angstroms.
20. The semiconductor structure of claim 14, wherein the thickness of the first dielectric layer and the thickness of the second dielectric layer are less than approximately 20 angstroms.
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